Lab 7 – Fuzz Testing

Daycare Fees

Name: Sruthi Vejju
SRN: PES1UG24CS472
Sec: 5H

Objective

The aim of this lab was to use Hypothesis to test the input handling of the Daycare Fees program. The program has three functions: parsing enrolment details, calculating average weekly hours and calculating the payable fee.

Tools used

Python 3.12, pytest and Hypothesis.

1. Running the original program

First, the original program was executed with normal sample inputs. The outputs were checked for all three functions.

```
$ python daycare_fees.py
==== Running parse_enrolment ====
Enter age group:hours (e.g. toddler:25): toddler:25
Age Group and Hours: ('Toddler', 25.0)

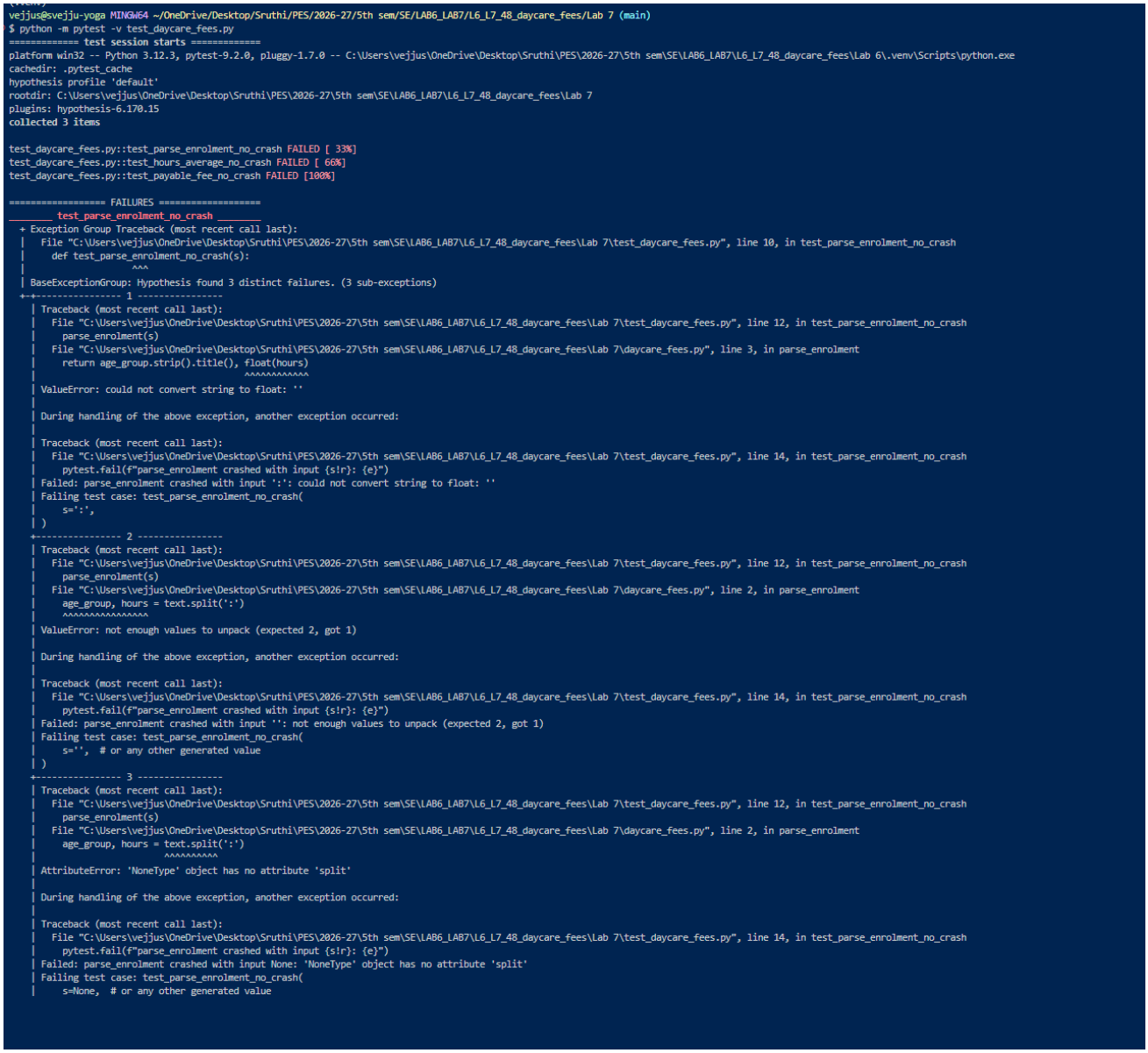
==== Running hours_average ====
Enter weekly hours as CSV (e.g. 20,25,30): 20,25,30
Average Hours: 25

==== Running payable_fee ====
Enter fee|discount (e.g. 3000|250): 3000|250
Payable Fee: 3290.0
(.venv)
```

2. Fuzz testing before fixing the code

Hypothesis generated different string inputs, including empty and invalid values. The initial tests found input-related errors in all three functions. Empty values, missing separators and None caused exceptions.

Initial fuzz-test failures

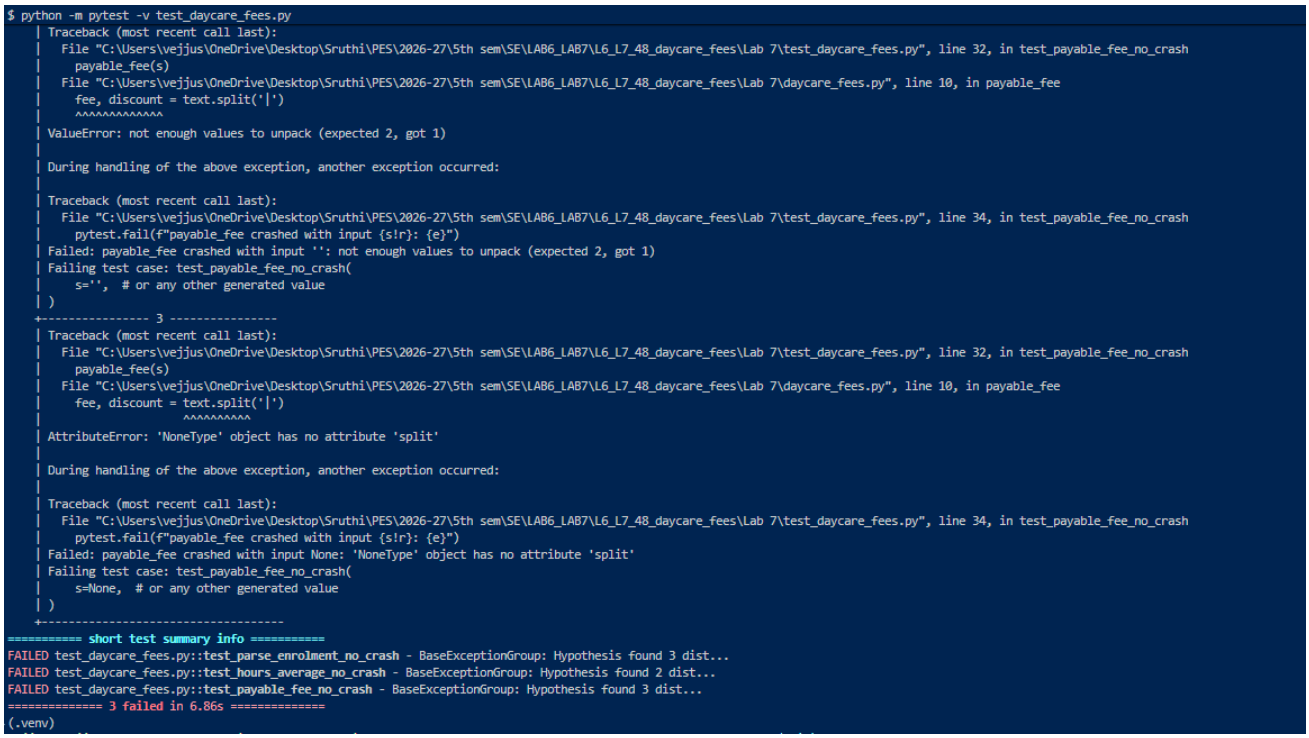


```
vejjus@svejju-yoga MINGW64 ~/OneDrive/Desktop/Sruthi/PES/2026-27/5th sem/SE/LAB6_LAB7/L6_L7_48_daycare_fees/Lab 7 (main)
$ python -m pytest -v test_daycare_fees.py
============ test session starts ============
platform win32 -- Python 3.12.3, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 6\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_daycare_fees.py::test_parse_enrolment_no_crash FAILED [ 33%]
test_daycare_fees.py::test_hours_average_no_crash FAILED [ 66%]
test_daycare_fees.py::test_payable_fee_no_crash FAILED [100%]

================ FAILURES ================
_______ test_parse_enrolment_no_crash _______
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 10, in test_parse_enrolment_no_crash
  |     def test_parse_enrolment_no_crash(s):
  |                ^^^
  | BaseExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 12, in test_parse_enrolment_no_crash
    |     parse_enrolment(s)
    |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\daycare_fees.py", line 3, in parse_enrolment
    |     return age_group.strip().title(), float(hours)
    |                              ^^^^^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 14, in test_parse_enrolment_no_crash
    |     pytest.fail(f"parse_enrolment crashed with input {s!r}: {e}")
    | Failed: parse_enrolment crashed with input ':': could not convert string to float: ''
    | Failing test case: test_parse_enrolment_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 12, in test_parse_enrolment_no_crash
    |     parse_enrolment(s)
    |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\daycare_fees.py", line 2, in parse_enrolment
    |     age_group, hours = text.split(':')
    |     ^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 14, in test_parse_enrolment_no_crash
    |     pytest.fail(f"parse_enrolment crashed with input {s!r}: {e}")
    | Failed: parse_enrolment crashed with input '': not enough values to unpack (expected 2, got 1)
    | Failing test case: test_parse_enrolment_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 12, in test_parse_enrolment_no_crash
    |     parse_enrolment(s)
    |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\daycare_fees.py", line 2, in parse_enrolment
    |     age_group, hours = text.split(':')
    |                        ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 14, in test_parse_enrolment_no_crash
    |     pytest.fail(f"parse_enrolment crashed with input {s!r}: {e}")
    | Failed: parse_enrolment crashed with input None: 'NoneType' object has no attribute 'split'
    | Failing test case: test_parse_enrolment_no_crash(
    |     s=None,  # or any other generated value
```

```
> python -m pytest -v test_daycare_fees.py
  | )
  +------------------------------------
_______ test_hours_average_no_crash ________
+ Exception Group Traceback (most recent call last):
|   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 20, in test_hours_average_no_crash
|     def test_hours_average_no_crash(s):
|                    ^^^
| BaseExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
+-+---------------- 1 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 22, in test_hours_average_no_crash
  |     hours_average(s)
  |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\daycare_fees.py", line 6, in hours_average
  |     values = [int(r) for r in hours_csv.split(',')]
  |               ^^^^^^
  | ValueError: invalid literal for int() with base 10: ''
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 24, in test_hours_average_no_crash
  |     pytest.fail(f"hours_average crashed with input {s!r}: {e}")
  | Failed: hours_average crashed with input '': invalid literal for int() with base 10: ''
  | Failing test case: test_hours_average_no_crash(
  |     s='',  # or any other generated value
  | )
  +---------------- 2 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 22, in test_hours_average_no_crash
  |     hours_average(s)
  |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\daycare_fees.py", line 6, in hours_average
  |     values = [int(r) for r in hours_csv.split(',')]
  |                              ^^^^^^^^^^^^^^
  | AttributeError: 'NoneType' object has no attribute 'split'
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 24, in test_hours_average_no_crash
  |     pytest.fail(f"hours_average crashed with input {s!r}: {e}")
  | Failed: hours_average crashed with input None: 'NoneType' object has no attribute 'split'
  | Failing test case: test_hours_average_no_crash(
  |     s=None,  # or any other generated value
  | )
  +------------------------------------
_______ test_payable_fee_no_crash _________
+ Exception Group Traceback (most recent call last):
|   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 30, in test_payable_fee_no_crash
|     def test_payable_fee_no_crash(s):
|                    ^^^
| BaseExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
+-+---------------- 1 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 32, in test_payable_fee_no_crash
  |     payable_fee(s)
  |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\daycare_fees.py", line 11, in payable_fee
  |     total = float(fee) * 1.18 - float(discount)
  |             ^^^^^^^^^
  | ValueError: could not convert string to float: ''
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 34, in test_payable_fee_no_crash
  |     pytest.fail(f"payable_fee crashed with input {s!r}: {e}")
  | Failed: payable_fee crashed with input '|': could not convert string to float: ''
  | Failing test case: test_payable_fee_no_crash(
  |     s='|',
  | )
  +---------------- 2 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 32, in test_payable_fee_no_crash
```



3. Changes made

Input validation was added to the three functions. The program checks that the input is a non-empty string, that the required separators are present, and that numeric values can be converted. It also checks for negative or non-finite values where applicable. Invalid input is reported using ValueError. The main program catches this error and displays an invalid-input message.

4. Running the updated program

After making the changes, the program was run again with the same sample inputs. The outputs were the same as the original run for these valid inputs.

```
vejjus@svejju-yoga MINGW64 ~/OneDrive/Desktop/Sruthi/PES/2026-27/5th sem/SE/LAB6_LAB7/L6_L7_48_daycare_fees/Lab 7 (main)
$ python daycare_fees.py
==== Running parse_enrolment ====
Enter age group:hours (e.g. toddler:25): toddler:25
Age Group and Hours: ('Toddler', 25.0)

==== Running hours_average ====
Enter weekly hours as CSV (e.g. 20,25,30): 20,25,30
Average Hours: 25

==== Running payable_fee ====
Enter fee|discount (e.g. 3000|250): 3000|250
Payable Fee: 3290.0
(.venv)
```

5. Fuzz tests after fixing the code

The tests were updated so that ValueError is treated as an expected response for invalid input. Other unexpected exceptions cause the test to fail. The three Hypothesis tests passed when run with pytest.

```
vejjus@svejju-yoga MINGW64 ~/OneDrive/Desktop/Sruthi/PES/2026-27/5th sem/SE/LAB6_LAB7/L6_L7_48_daycare_fees/Lab 7 (main)
$ python -m pytest -v test_daycare_fees.py
============================================================ test session starts ============================================================
platform win32 -- Python 3.12.3, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 6\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\vejjus\OneDrive\Desktop\Sruthi\PES\2026-27\5th sem\SE\LAB6_LAB7\L6_L7_48_daycare_fees\Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_daycare_fees.py::test_parse_enrolment_no_unexpected_crash PASSED                                                                [ 33%]
test_daycare_fees.py::test_hours_average_no_unexpected_crash PASSED                                                                  [ 66%]
test_daycare_fees.py::test_payable_fee_no_unexpected_crash PASSED                                                                    [100%]

============================================================ 3 passed in 1.46s ============================================================
(.venv)
```

Result

The original fuzz tests exposed input validation problems. After adding validation and updating the tests to accept expected ValueError responses, all 3 fuzz tests passed. The valid sample calculations continued to produce the expected outputs.